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Finite element analysis with staggered gradient elasticity

Harm Askes [a,*], Irene Morata [b], Elias C. Aifantis [c,d]

[a] University of Sheffield, Department of Civil and Structural Engineering, Mappin Street, Sheffield S1 3JD, United Kingdom
[b] Universitat Politècnica de Catalunya, Laboratori de Càlcul Numèric, Barcelona, Spain
[c] Aristotle University of Thessaloniki, Polytechnic School, Thessaloniki, Greece
[d] Michigan Technological University, Center for Mechanics of Materials and Instabilities, Houghton, USA

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Abstract

In this article, staggered gradient elasticity formulations are studied. Firstly, the standard equations of classical elasticity are considered. Afterwards, a set of Helmholtz equations associated with the theory of gradient elasticity is solved to handle the gradient dependence. Due to the two-step nature of the algorithms, $\mathscr{C}^0$-continuous interpolation functions suffice and finite element discretisations are straightforward and efficient. Different versions of staggered gradient elasticity are treated, whereby the Helmholtz equations operate on the displacements, on the strain tensor, on the stress tensor or on a strain invariant. The governing equations are given with their consistent boundary conditions. The formulations are tested against two criteria: whether all singularities are removed from the strain field, and whether the models are capable of describing size effects.
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1. Introduction

Higher-order gradients of strain and state variables are commonly introduced in the continuum description in order to conveniently account phenomenologically for the effects of microstructure on the macroscopic response of materials and structures. This has been recognised for elasticity in the classical works of Toupin [1] and Mindlin [2] who have cast earlier proposals of Bernoulli, Euler, Cauchy, Voigt and the Cosserat brothers, as well as of other more recent workers of the 20th century, into modern theories of generalised continua. A brief account of these works in relation to the gradient elasticity model to be analysed here is provided in the review by Altan and Aifantis [3]. A more recent account of gradient theory as applied to elasticity, plasticity and dislocation dynamics can be found in an article by Aifantis [4] where various important implications such as in the elimination of elastic singularities, on the prediction of size effects in elasticity and plasticity, as well as on the description of dislocation patterning phenomena are reviewed. More details on the role of higher-order strain gradients in dissipative materials can be found in [5,6] where the first robust gradient plasticity theory was proposed, whereas an improved (deformation type) version of it also accounting for interfacial energy effects was recently published in [7]. The removal of elastic strain singularities from dislocation lines and crack tips is discussed in detail in [8–14]. Moreover, wave dispersion in dynamics can be described realistically if gradients of inertia are included [15–19]. Similarly, if the higher-order gradients are included in a suitable manner, strain-softening problems are regularised and mesh-objective numerical solutions can be obtained [20–28]. Finally, as already mentioned, gradient-enriched continua may be used to describe size effects. The higher-order gradients are accompanied by additional material coefficients in terms of one or more internal length scales. The interaction between the material length and the structural dimensions leads to the occurence of size effects, see for instance [8,29–38].

* Corresponding author.
E-mail address: h.askes@sheffield.ac.uk (H. Askes).

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An important issue, if numerical simulations are to be carried out with gradient models, concerns the requirements imposed on the discretisation. The standard equations of solids mechanics (elasticity, plasticity, damage), which are commonly of second-order, can be discretised with $\mathscr{C}^0$-continuous shape functions. In contrast, the governing equations of gradient models may necessitate shape functions that are $\mathscr{C}^1$-continuous or even higher. In the most popular discretisation method of solid mechanics, namely the finite element method, $\mathscr{C}^1$ shape functions are by no means straightfowardly formulated and implemented. Indeed, the additional complexity as compared to $\mathscr{C}^0$ shape functions is considerable. Different strategies have been taken in the literature to implement gradient models. The following overview is not meant to be complete, but rather to give a flavour of the various directions of research, with an emphasis on Galerkin methods:

Strongly $\mathscr{C}^1$ Galerkin methods:

- Hermitian finite elements may be formulated in isoparametric space [39], thus avoiding earlier restrictions of rectangular domains. However, an important increase in nodal variables is common. Applications towards gradient plasticity models include [26,40,41].
- Meshless methods such as the element-free Galerkin method or the Meshless Local Petrov–Galerkin method lead to shape functions that are easily formulated with an arbitrary order of continuity. Thus, the increase in nodal unknowns is avoided, and implementation of gradient models is straightforward [31,42–46], even if $\mathscr{C}^2$-continuity is required [47–49]. However, the computational cost of meshless methods is in general higher than that of finite elements, and the interaction between nodes is normally extended beyond the nearest neighbours.
- For finite element methods, the condition can be dropped that the displacement field within an element only depends on the nodal unknowns of that particular elements. Then, higher-order continuity can be obtained without resorting to higher-order derivatives as unknowns. Instead, *next nearest neighbour* (rather than *nearest neighbour*) interaction is introduced between the nodes. This approach is known as the subdivision surfaces paradigm [50].

Remark 1. A similar kind of next nearest neighbour interaction has been used in the *formulation* (rather than implementation) of gradient theories [51]. Long-range and short-range interactions between particles were used in Taylor-series expansions to arrive at a continuum model extended with higher-order gradients of the strain.

Weakly $\mathscr{C}^1$ Galerkin methods:

- Penalty methods can be used to enforce the higher-order continuity requirements by satisfying the relation between the various kinematic quantities in a weak (averaged) sense. Care must be taken that any gradient-enhanced failure condition and its consistency condition are satisfied pointwise, and sub-optimal integration points can be necessary [26].
- Equally, Lagrange multipliers can be used to relate the various kinematic fields in a weak sense, by which so-called mixed finite elements are obtained [52–54]. Also, the concept of discrete Kirchhoff constraints can be used [55]. Normally, the number of degrees-of-freedom per element increases importantly.
- In continuous/discontinous Galerkin methods the use of additional nodal unknowns and higher-order continuity of the shape functions is avoided. The kinematic relations are weakly enforced at interelement boundaries through stabilisation terms, whereby next nearest neighbour interaction is introduced between the nodes [56,57]. It was shown in [57] that the one-dimensional case with uniform finite element mesh yields equations that are identical to the corresponding finite difference equations.

All the above approaches leave the continuum mechanical equations intact. Alternatively, the mechanical equations could be altered such that less strict continuity requirements have to be met. For instance, in [8] the original fourth-order partial differential equations of gradient elasticity have been rewritten as two sets of second-order partial differential equations, which can then be solved consecutively. A finite element implementation of the two-step method was presented in [58], although different boundary conditions were taken compared to [8].

A second example of rewriting the mechanical equations to lower the continuity requirements concerns the work presented in [27,59]. Two families of gradient-enhanced damage models have been derived from an integral non-local damage model. The second-order models of both families require $\mathscr{C}^1$ and $\mathscr{C}^0$-continuity. It was found that the latter not only has less stringent continuity requirements, but it also shows superior mechanical behaviour [59,60]. This approach has recently also been extended to elasticity [31,61]. Even though not always explicitly mentioned in the literature, this class of models result from the same non-local Taylor expansion used to derive an extra strain in gradient plasticity theory (e.g. [23,24]). This result is different from the case of gradient elasticity where the inhomogeneous Helmholtz equation is derived from the equilibrium equation, the constitutive equation as well as the strain–displacement relations, rather than from the constitutive equations alone.

The above two approaches have in common that an inhomogeneous Helmholtz equation is solved as a post-processing step after solving the classical mechanical equations. Due to the two-step nature of these approaches they are referred to as *staggered*. The classical mechanical equations yield the so-called *local* displacements and the *local* strains. In the subsequent Helmholtz equations the *non-local* variable is the unknown whilst the local variable acts as a source term. In [8,58], the Helmholtz equation is formulated in

terms of displacements and it is solved separately from the classical equations of elasticity. In [27] and follow-up works, the Helmholtz equation is formulated in terms of an equivalent strain and it is solved simultaneously with the classical equations of elasticity-based damage. However, prior to damage initiation, the two sets of equations in [27] are uncoupled, and in the elastic regime the Helmholtz equation can again be used *after* solving the classical equations of elasticity. Recently, an alternative non-local damage model has been formulated where a Helmholtz equation is used for the displacements, which are then used to compute non-local strains and the damage evolution [62].

In this contribution, alternative staggered gradient elasticity formulations are proposed. The ensuing Helmholtz equation operates on the strain tensor or on the stress tensor (rather than the displacement vector or a strain invariant) and can formally be derived from the displacement-based approach. A rigorous comparison is carried out between the aforementioned two staggered gradient elasticity formulations and two newly proposed ones. The strong forms of the formulations are presented in a unified manner in Section 2. The weak form and the variationally consistent boundary conditions are discussed in Section 3, where also a simple one-dimensional example is treated to illustrate the qualitative and quantitative effects of the boundary conditions. Section 4 treats some issues regarding the finite element implementation. In Sections 5 and 6 the performance of the approaches in two sets of applications is dealt with, namely in the removal of singularities from the strain field (Section 5) and the description of size effects (Section 6).

2. Gradient elasticity – strong forms

In the simplest form of gradient elasticity, as suggested by Aifantis and coworkers (e.g. [3,4,8,9,11,22]), the constitutive equations are written as

$$\sigma_{ij} = C_{ijkl}(\varepsilon_{kl} - \ell^2 \varepsilon_{kl,mm}) \tag{1}$$

in which σ is the Cauchy stress, ε is the usual infinitesimal strain, C contains the elastic moduli and ℓ is a material length parameter that represents the underlying microstructure. Indices following a comma denote a spatial derivative.

The newly introduced parameter ℓ can be related to microstructural properties: for a regular lattice of discrete particles and springs, the parameter ℓ is related to the inter-particle distance [17], and for a heterogeneous material consisting of aggregates randomly distributed in a matrix material the parameter ℓ can be linked one-to-one with the dimensions of the representative volume element via homogenisation procedures [63–65]. The format as given in Eq. (1) has proven to be effective in statics, in particular in the removal of singularities from the strain field [3,6,8–14,18,48,66]. However, for the realistic description of dispersive effects in dynamics other amendments may be necessary; in particular, the notion of higher-order inertia seems powerful as was suggested from a phenomenological point of view [22,67] and derived from microstructural arguments [15–19]. The negative sign that precedes the higher-order term in Eq. (1) is dictated by arguments of stability and uniqueness [3,20,48]. In the literature it has sometimes been suggested to use a positive sign; in [48,68–70] the two approaches and their behaviour in statics and dynamics are compared.

The constitutive relations given by Eq. (1) are combined with the equilibrium equations

$$\sigma_{ij,j} = 0 \tag{2}$$

in the absence of body forces, and the kinematic equations

$$\varepsilon_{kl} = \frac{1}{2}(u_{k,l} + u_{l,k}) \tag{3}$$

to derive a system of equations that is fourth-order in the displacements u, i.e.

$$\frac{1}{2} C_{ijkl}(u_{k,jl} + u_{l,jk} - \ell^2 (u_{k,jl} + u_{l,jk})_{,mm}) = 0, \tag{4}$$

whereby homogeneous material is assumed. If a numerical solution technique is chosen, $\mathscr{C}^1$-continuous shape functions are needed, such as for instance those provided by meshless methods [31,44]. However, system (4) can be rewritten by means of the Ru–Aifantis theorem [8] such that $\mathscr{C}^0$-continuous shape functions suffice.

2.1. The Ru–Aifantis theorem in terms of displacements

It was noted in [8] that Eq. (4) can be rewritten by changing the order of the various derivatives. In particular,

$$\frac{1}{2} C_{ijkl}((u_k - \ell^2 u_{k,mm})_{,jl} + (u_l - \ell^2 u_{l,mm})_{,jk}) = 0. \tag{5}$$

The terms between the inner parentheses in Eq. (5) can be identified as the components of another displacement field u^c that obeys the classical equations of elasticity:

$$\frac{1}{2} C_{ijkl}(u^c_{k,jl} + u^c_{l,jk}) = 0. \tag{6}$$

The relation between the gradient-enriched displacements u^g and the classical displacements u^c then follows from a set of non-homogeneous Helmholtz equations:

$$u^g_i - \ell^2 u^g_{i,jj} = u^c_i. \tag{7}$$

Note that u^g is identical to u in Eqs. (4) and (5) – the superscript g has been introduced to distinguish the gradient-enriched (or non-local in a somewhat loose terminology) displacements u^g from its classical (or local) counterpart u^c. This notation will be used in the remainder of the paper.

Eqs. (6) and (7) replace the original governing equations of gradient elasticity given by Eq. (4). The obvious advantage is that once the solution of classical elasticity is known, the complexity of solving Eq. (7) is reduced as compared to solving the original fourth-order Eq. (4). This has been explored using analytical solution strategies in [8,68,69], among others. However, also in a numerical context Eqs. (6) and (7) have a clear potential: $\mathscr{C}^0$-continuous

shape functions suffice for the interpolation of both u^c and u^g; for instance, see [58].

Since the governing equations underlying Eqs. (6) and (7) on the one hand and Eq. (4) on the other hand are intrinsically identical, the only difference between them could be generated by using different boundary conditions. In [8] the additional boundary conditions were specified for the normal derivatives of the strain (that is, the second derivatives of the displacement). However, within a finite element context the variationally consistent format of the natural boundary conditions is to take the derivatives of the displacement, not of the strain; see, for instance, Section 3 and [58].

2.2. Strain-based Ru–Aifantis theorem

In order to bring the variationally consistent boundary conditions in accordance with those used in [8], a modified Ru–Aifantis theorem is suggested here. Eq. (7) is translated into terms of strain by simply taking derivatives [12,13], i.e.

$$\varepsilon^g_{ij} - \ell^2 \varepsilon^g_{ij,mm} = \varepsilon^c_{ij}, \tag{8}$$

which relates the gradient (or non-local) strain tensor to the classical (or local) strain tensor. This process requires that the displacement fields are sufficiently smooth so that the derivation of Eq. (8) from Eq. (7) is, indeed, meaningful. Both strain tensors are expressed in terms of the corresponding displacements via Eq. (3). This approach has also been suggested in the context of damage mechanics in [71].

The staggered approach now consists of solving the equations of classical elasticity, i.e. Eq. (6), computing the local strains ε^c and using this quantity as a source term for the solving of the Helmholtz equation, i.e. Eq. (8). Eqs. (7) and (8) are equivalent if the material length scale ℓ is uniform throughout the domain under consideration and appropriate smoothness assumptions are adopted. Note that the unknown in Eq. (8) is the non-local strain, not the non-local displacement. The essential boundary conditions in terms of displacements as suggested in [8] are therefore irrelevant in this approach.

2.3. Stress-based Ru–Aifantis theorem

The stress fields obtained with classical elasticity via Eq. (6) or with gradient elasticity via Eq. (4) are identical (and will here be denoted as σ^c). This implies that if ε^c is singular, then from Eqs. (8) and (1) it follows that σ^c is singular as well. To overcome this, it has been suggested to use σ^c in another Helmholtz equation as [12,13,4]

$$\sigma^g_{ij} - \ell^2 \sigma^g_{ij,mm} = \sigma^c_{ij}. \tag{9}$$

If the two stress tensors are both identified as following Hooke's law with $\sigma^g_{ij} = C_{ijkl}\varepsilon^g_{kl}$ and $\sigma^c = C_{ijkl}\varepsilon^c_{kl}$, then it is easily seen that Eq. (9) follows from Eq. (8) by simple pre-multiplying with the constitutive tensor C. Using Eq. (8) or Eq. (9) has implications for the format of the variationally consistent boundary conditions. Whereas the strain-based Ru–Aifantis theorem leads to boundary conditions that are in line with the ones suggested in [8], the stress-based Ru–Aifantis theorem renders boundary conditions that correspond to those of the original fourth-order equation. This will be discussed in detail in Section 3.

2.4. Peerlings' gradient damage

A strategy similar to Eq. (8) has been used in the context of gradient damage by Peerlings and coworkers, see for instance [27,59,60]. Instead of the strain tensor, an equivalent strain measure is used in the Helmholtz equation:

$$\varepsilon^g_{eq} - \ell^2 \varepsilon^g_{eq,mm} = \varepsilon^c_{eq}. \tag{10}$$

The equivalent strain ε_{eq} is a function of the strain invariants, which could be taken as the Mazars equivalent strain [72], that is

$$\varepsilon_{eq} = \sqrt{\sum_{i=1}^{3} \langle e_i \rangle^2} \tag{11}$$

with $\langle e_i \rangle = \frac{1}{2} e_i + \frac{1}{2}|e_i|$ and e_i the ith principal strain. The equivalent strain measure serves as an indicator of failure initiation. In the original works on gradient damage [27,60], Eq. (10) is solved *simultaneously* with the equations of elasto-damage. The coupling terms that connect the two sets of equations are related to damage growth. However, in the elastic regime Eq. (10) can be solved separately from the classical elasto-damage equations, and the treatment of Eq. (10) follows similar lines as that of Eqs. (7) and (8).

3. Weak forms and boundary conditions

The staggered gradient elasticity approaches of Eqs. (7)–(9) will be denoted as u-RA, ε-RA and σ-RA, respectively. For a unified treatment of the staggered approaches, it is noted that if the Helmholtz equation is applied to a vector or on a tensor, it is applied to each component separately. Thus, there is no fundamental difference in the treatment of Eqs. (7)–(10). The weak form of the Helmholtz equation in terms of a generic variable s is written as

$$\int_\Omega \delta s (s^g - \ell^2 s^g_{,ii}) \mathrm{d}\Omega = \int_\Omega \delta s s^c \mathrm{d}\Omega, \tag{12}$$

where Ω is the domain under consideration, s represents a displacement component, a strain component, a stress component or an equivalent strain, and δs is a corresponding test function. Integration by parts of the second-order term yields

$$\int_\Omega \delta s\, s^g \mathrm{d}\Omega + \int_\Omega \delta s_{,i}\, \ell^2\, s^g_{,i} \mathrm{d}\Omega = \int_\Omega \delta s\, s^c \mathrm{d}\Omega + \oint_\Gamma \delta s\, \ell^2\, n_i s^g_{,i} \mathrm{d}\Gamma, \tag{13}$$

where Γ is the boundary of Ω and n is the outward normal to Γ. The two integrals on the left-hand side of Eq. (13) are

cast in terms of the unknown variable s^g. The first integral on the right-hand side acts as a source in terms of the known variable s^c. As is seen from the last integral in Eq. (13), the natural boundary conditions are written in terms of first derivatives of the non-local quantity s^g. Thus, if the Helmholtz equation is applied to displacements, displacement derivatives should be prescribed as natural boundary conditions, and if the Helmholtz equations is applied to strains, the derivatives of the strains should be prescribed as natural boundary conditions.

3.1. Discussion of boundary conditions

The boundary conditions derived above should be contrasted with those valid for the original fourth-order format of gradient elasticity, cf. Eq. (1). The variationally consistent boundary conditions for gradient elasticity according to Eq. (1) have been discussed in a number of recent articles, e.g. [73–75]. The *essential boundary conditions* are cast in terms of displacements as well as their normal derivatives, whereas the *natural boundary conditions* are the conjugated tractions, i.e. standard tractions t and a higher-order tractions m. As usual, the essential boundary conditions concern kinematic (displacement type) variables and the natural boundary conditions concern dynamic (stress type) variables. Specifically, the essential boundary conditions of the fourth-order model are expressed in terms of the displacement, i.e.

$$u_i = \bar{u}_i \tag{14}$$

for the standard essential boundary conditions, and in terms of the normal derivative, i.e.

$$\frac{\partial u_i}{\partial x_j} n_j = \bar{\eta}_i \tag{15}$$

for the higher-order essential boundary conditions [52]. Here, $\bar{u}$ and $\bar{\eta}$ denote prescribed values for the displacement and for the normal derivative of the displacement, respectively. The *standard* tractions t (in terms of Cauchy stress σ as given in Eq. (1), the higher-order stress τ and the normal vector n) take a relatively complicated form [52,73]:

$$t_i = n_j\left(\sigma_{ij} + n_k\tau_{ijk}(\delta_{lm} - n_l n_m)\frac{\partial n_l}{\partial x_m}\right) - (\delta_{km} - n_k n_m)\frac{\partial n_j \tau_{ijk}}{\partial x_m} \tag{16}$$

with the higher-order stresses $\tau_{ijm} = \ell^2 C_{ijkl}\varepsilon^g_{kl,m}$. The *higher-order* tractions m are expressed in terms of the higher-order stress τ and the normal vector n as

$$m_i = n_j n_k \tau_{ijk}. \tag{17}$$

Thus, the higher-order tractions are related to the strain derivatives, i.e. to the second derivatives of the displacements. Natural boundary conditions are then prescribed by specifying values $\bar{t}$ and $\bar{m}$ such that $t_i = \bar{t}_i$ and $m_i = \bar{m}_i$.

The field equations solved in the *staggered approaches of gradient elasticity* are different from those of the original fourth-order equations of gradient elasticity. Hence, the boundary conditions will also be different. In Table 1 all boundary conditions of the various formulations have been summarised. The first and second row deal with the standard boundary conditions, i.e. displacement and standard tractions; in the third and fourth row the higher-order boundary conditions are given. For the displacement-based Ru–Aifantis formulation a distinction is made between the strong form given in [8] and the weak form used for finite element implementation in [58]. Since the various staggered approaches all use Eq. (6) as a first step, they do not differ as concerns the standard boundary conditions in terms of displacements and standard tractions (first and second row of Table 1). Therefore attention is focussed on the higher-order boundary conditions (third and fourth row):

1. The essential boundary condition for the Peerlings model in terms of ε^g_{eq} is merely stated for reasons of completeness – in the literature the corresponding natural boundary condition is used throughout.
2. In the original publication suggesting the Helmholtz approach to displacements [8], strain derivatives (not displacement derivatives) were used as the higher-order natural boundary conditions. If the strong form of u-RA is used, the analyst is more or less free to select boundary conditions, and strain derivatives can be used as demonstrated in [8,68,69]. However, if the weak form of u-RA is used, displacement derivatives are dictated by the weak form. Indeed, the natural boundary conditions are cast as strain derivatives and displacement derivatives in [8] and [58], respectively.
3. The original equations of gradient elasticity, that is Eq. (4), lead to two natural boundary conditions and two essential boundary conditions which are of the dynamic type and of the kinematic type, respectively. In contrast, the u-RA, ε-RA and Peerlings approaches all render one dynamic boundary condition (namely the standard natural boundary condition in terms of standard tractions) and three kinematic boundary conditions (i.e. the standard essential boundary condition as well as both higher-order boundary conditions). On the other hand, the σ-RA approach yields an additional natural boundary condition that is of the dynamic type, similar to Eq. (17).
4. The interpretation of displacement derivatives as natural boundary conditions is cumbersome for traction boundaries. Due to the kinematic character of this boundary

Table 1
Overview of boundary conditions

	Original (fourth-order)	u-RA as in [8]	u-RA (weak form)	ε-RA	σ-RA	Peerlings
Essential	u^g	u^c	u^c	u^c	u^c	u^c
Natural	t	t	t	t	t	t
Essential	$\partial u^g/\partial n$	u^g	u^g	ε^g	σ^g	ε^g_{eq}
Natural	m	$\partial^2 u^g/\partial n^2$	$\partial u^g/\partial n$	$\partial\varepsilon^g/\partial n$	$\partial\sigma^g/\partial n$	$\partial\varepsilon^g_{eq}/\partial n$

condition, interpretation of non-zero values is not clear as yet. Conversely, zero values may lead to severe deviations from the classical results where this is not expected, as for instance shown in [58].

In the strong form of the u-RA approach, the higher-order boundary conditions are taken as homogeneous natural boundary conditions [8,68,69]. In the ε-RA approach, the σ-RA approach and the Peerlings approach, this is easily reproduced by taking homogeneous higher-order natural boundary conditions – this will be used in the finite element implementation and the subsequent numerical examples discussed below. In the weak form of the u-RA approach, on the other hand, zero higher-order tractions cannot be imposed directly. The Helmholtz Eq. (7) has been used with essential boundary conditions $u_i^{\mathrm{g}} = u_i^{\mathrm{c}}$ whereever the classical displacements u_i^{c} are prescribed, and with natural boundary conditions $u_{i,j}^{\mathrm{g}} n_j = 0$ elsewhere.

3.2. Illustration: one-dimensional bar problem

The differences between the various formulations become relevant in case of stiffness jumps, whereby at the interfaces all four essential and natural boundary conditions must be specified. This is best illustrated by studying an inhomogeneous bar example as shown in Fig. 1. The bar consists of two pieces with different Young's modulus, E_1 and E_2 respectively, while the length parameter ℓ is the same for both materials. All governing equations (field equations, boundary conditions and interface conditions) are treated in detail in Appendix B. For this one-dimensional case there is no difference between the ε-RA approach and the Peerlings approach, hence only the ε-RA approach is treated.

In Fig. 2 the axial strain ε_{xx} is plotted taking $E_1/E_2 = 5$, $L = 10\,\mathrm{mm}$, $\ell = 2\,\mathrm{mm}$ and $\bar{u} = 0.1\,\mathrm{mm}$. The original fourth-order approach shows a jump in slope that corresponds to the jump in stiffness. The u-RA (weak form) and ε-RA approaches yield identical results, as dictated by the boundary conditions and interface conditions. However, they differ from the original approach in that no jump in slope is present. This difference is the result of the difference in higher-order natural boundary conditions: in the fourth-order model the higher-order tractions m include the effects of stiffness, in contrast to the natural boundary conditions of the u-RA and ε-RA approaches. On the other hand, the stiffness jumps are reflected in the interface conditions for the σ-RA approach, and this model leads to results that differ only quantitatively (not qualitatively) from those of the fourth-order model.

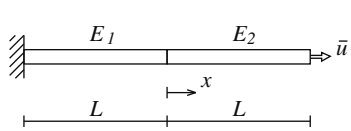

Fig. 1. Bar with discontinuous material properties.

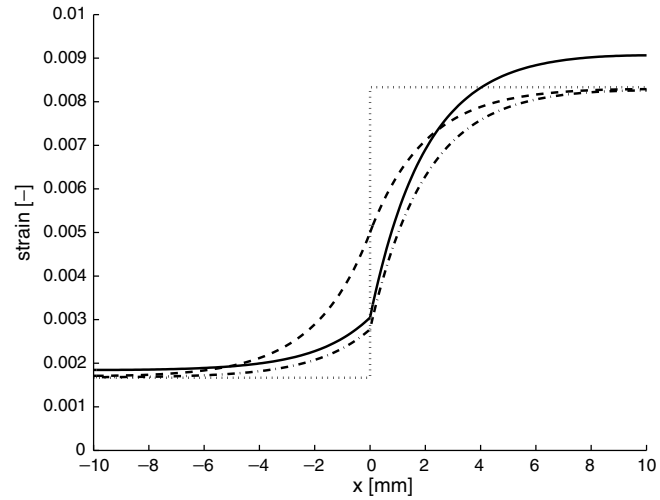

Fig. 2. Axial strain ε_{xx} versus x for the bar problem of Fig. 1: response of the original fourth-order approach (solid), the u-RA and ε-RA approaches (dashed), the σ-RA approach (dot-dashed) and classical elasticity (dotted).

As is explained in Appendix B, the strong form of the u-RA approach cannot be analysed for this configuration. For comparison reasons the response of classical elasticity is plotted as well, by which the influence of the higher-order gradients in both approaches (original and staggered) can be verified.

4. Implementational issues

This section addresses some specific issues related to the finite element implementation of the three staggered gradient elasticity approaches. More details on implementation of the u-RA approach for a linear triangle may also be found in [58].

Interpolation functions: Since the two steps in the staggered approaches are solved separately, the interpolation functions can be chosen according to the user's preference. However, the output of the first step, Eq. (6), serves as input for the subsequent Helmholtz equation, so for efficiency reasons the accuracy in solving the two steps must be balanced. An obvious choice would be to use the same finite element mesh for both steps. In particular, the same element (bi-linear four-noded quadrilateral) is used for both steps in this study.

Component-wise decoupling: The Helmholtz equation can be solved component-wise, that is the equations of the various components of u^{g}, ε^{g} and σ^{g} are decoupled. Thus, if the same interpolation functions and material parameters are used in every spatial direction (as would be the obvious choice), then the system matrix is the same for every component. An efficient solution algorithm could be devised in which this system matrix is factorised only once and then used for every non-local component.

Post-processing: In the ε-RA, σ-RA and Peerlings approaches, the non-local strains and stresses are nodal variables, therefore no post-processing is needed for these

quantities. In contrast, the non-local strains are obtained as integration point variables in the u-RA approach. For a fair comparison between strains and stresses in all approaches, some post-processing is needed to arrive at nodal values for the non-local strains or stresses in the u-RA approach.

5. Numerical tests (1) – removal of singularities

Firstly, it will be verified whether singularities, that may be present in the solution of the equations of classical elasticity, are removed in the staggered approaches. To this end a strip with a crack is considered. For reasons of symmetry, only the top right quarter is modelled, and loading is induced via prescribed displacements, see Fig. 3 (top left). The magnitude of the imposed displacement is $u = L/100$ mm. The specimen dimensions are given as $L = 1$ mm which implies a crack length of 0.2 mm. The material constants read $E = 1000$ MPa and $\nu = 0$, whereas $\ell = 0.1$ mm. Symmetry is imposed by setting $u_i^g = 0$ for the u-RA approach and $\partial\varepsilon_{ij}^g/\partial n = 0$ for the ε-RA approach (see [41] for a detailed discussion of symmetry conditions in gradient theories). Due to the complexity of finite element implementations for the original fourth-order model no comparisons with this model are available. Some analytical results have been obtained for an infinite domain and they show that singularities are removed using the strong form of u-RA [8].

Remark 2. Since this test does not involve any inhomogeneity in terms of material stiffness, the field equations and boundary conditions of the ε-RA approach and the σ-RA approach are identical, by which their results also become identical. For this reason results for the σ-RA approach are not mentioned explicitly below.

For the finite element solutions of the staggered formulations, the occurence or absence of singularities is verified by performing a mesh refinement study: structured meshes of 5×5 up to 80×80 quadrilateral elements are used. It is well known that classical elasticity predicts singularities in the strain field at the crack tip. Firstly, the non-local strains obtained with the u-RA approach are investigated, see Fig. 3 (top right, centre left and centre right). The profiles of the strains along the line $y = 0$ are plotted. Upon mesh refinement, ε_{xx}^g converges towards a unique solution and no singularities are detected. In contrast, no convergence upon mesh refinement is found for ε_{yy}^g and γ_{xy}^g: unbounded peaks are found at the crack tip. Thus, singularities remain present in these two strain components. Next, the ε-RA approach and the Peerlings approach are studied. The non-local equivalent strain of these two approaches is plotted in Fig. 3 (bottom left and bottom right). Upon mesh refinement, a unique solution is obtained, and it is concluded that the singularities have been removed. Thus, the appropriate boundary conditions to be used are normal derivatives of the strain, not normal derivatives of the displacement.

Remark 3. Although not plotted here, no singularities were found for the individual non-local strain components in the ε-RA approach.

Remark 4. As can be seen in Fig. 3 (top right, bottom left and bottom right), the staggered schemes tend to shift the maximum of the strain components away from the crack tip. This has been analysed in detail for the Peerlings approach in [76] and also been quantified in [12].

Remark 5. The minor differences between the non-local equivalent strain of the ε-RA approach and the Peerlings approach are due to the non-linear character of Eq. (11).

6. Numerical tests (2) – description of size effects

Next, the ability of the formulations to describe size effects are studied. All engineering materials exhibit size effects, that is the mechanical response of a specimen depends on the actual dimensions of the specimen. It has long been recognised that size effects cannot be described by classical elasticity formulations. In contrast, gradient-enhanced continuum formulations are capable to capture size effects. In a previous contribution, a meshless implementation of the original fourth-order equations of gradient elasticity was presented and it was shown that size effects can be described [31]. Below, this will be extended to the staggered finite element implementations while also different configurations will be tested. In all tests, the material data read $E = 1000$ MPa, $\nu = 0.25$ and $\ell = 1$ mm, and plane strain conditions are assumed. The considered specimens are again homogeneous in terms of material stiffness, therefore the σ-RA approach is not treated.

6.1. Self-similar strip with circular hole

Firstly, the example of a strip with a circular hole is studied as discussed in [31]. The geometry and loading conditions are given in Fig. 4 (left). For reasons of symmetry, only the top right quarter of the strip is modelled. A nominal strain is present in the strip, the magnitude of which can be expressed as $\varepsilon_{\text{nom}} = \bar{u}/4R$, where $\bar{u}$ and R are the imposed displacement and the hole radius, respectively. The occurence of size effects is verified by investigating a strain ratio defined as

$$\text{strain ratio} = \frac{\varepsilon_{\text{nom}}}{\max(\varepsilon_{\text{eq}}^g)}. \tag{18}$$

The maximum non-local equivalent strain is obtained numerically and it occurs directly right (and left) of the hole. In Fig. 4 (right) the strain ratio is plotted against the ratio R/ℓ in a double logarithmic scale. The geometry dimensions that have been simulated are given through values $R = [1, 2, 4, 8, 16, 32, 64]$ mm.

The local equivalent strain according to classical elasticity does not exhibit a size effect, which is due to the absence

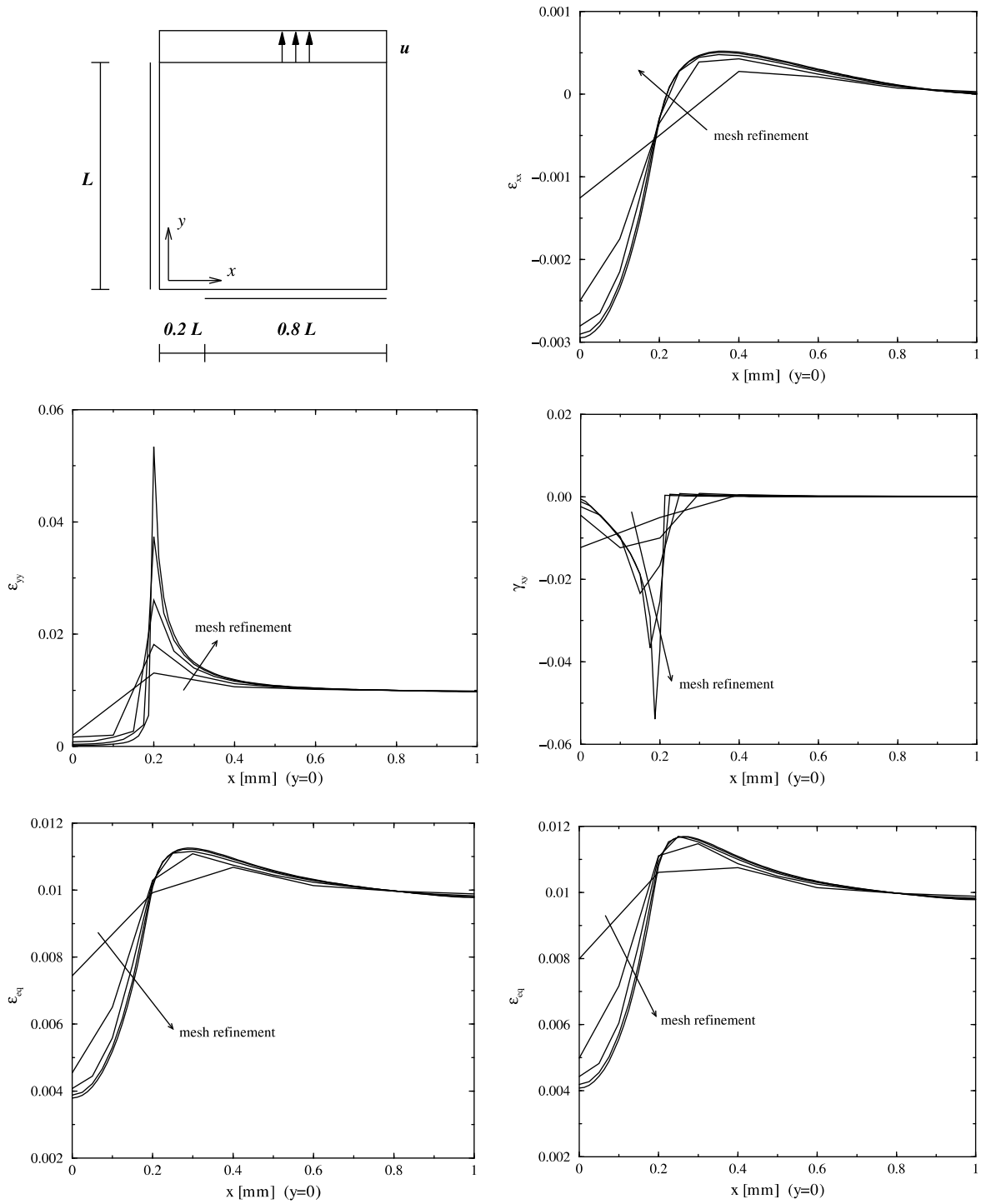

Fig. 3. Removal of singularities – geometry and loading conditions (top left), nonlocal strain components of u-RA approach ε^{g}_{xx} (top right), ε^{g}_{yy} (centre left) and γ^{g}_{xy} (centre right), nonlocal equivalent strain of ε-RA approach (bottom left) and Peerlings approach (bottom right).

of a material length scale. In contrast, the three staggered gradient elasticity formulations show an increasing strain ratio for decreasing specimen dimensions. In other words, the non-local equivalent strain decreases compared to the nominal strain if the specimen dimensions decrease compared to the material length scale. The size effects predicted by the ε-RA approach and the Peerlings approach are virtually identical, whereas the size effect predicted by the u-RA approach is significantly weaker.

6.2. Self-similar strip with ellipsoidal hole

In the second test the circular hole of Fig. 4 is replaced by an ellipsoidal hole. The semi-axes of the ellipse are

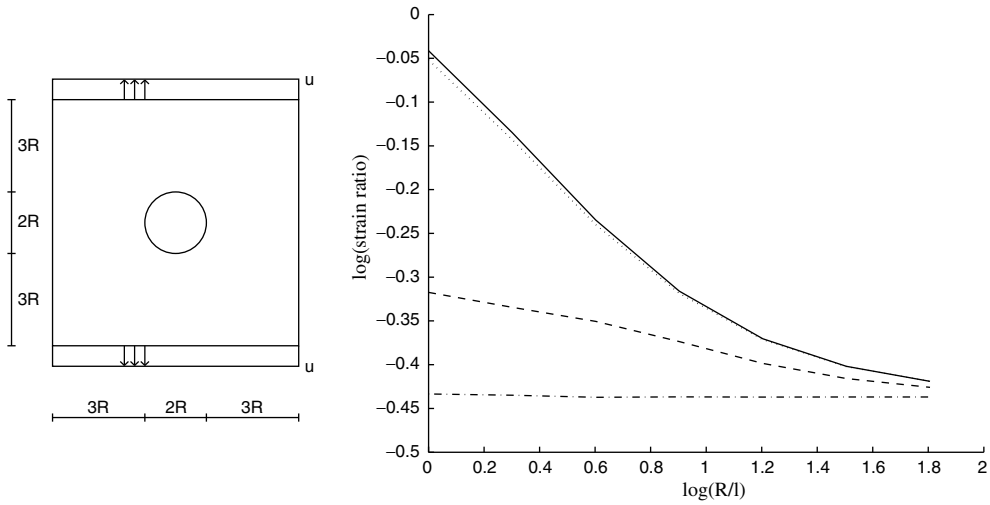

Fig. 4. Strip with circular hole – geometry and loading conditions (left) and size effect curves (right): u-RA approach (dashed), ε-RA approach (solid), Peerlings approach (dotted) and classical elasticity (dot-dashed).

denoted by a and b; with the origin of the coordinate axes centered at the ellipse, the ellipse is described by $x^2/a^2 + y^2/b^2 = 1$. Again, only the top right quarter of the specimen is modelled, and the strain ratio given in Eq. (18) is analysed. The width of the quarter specimen was taken equal to four times the horizontal ellipse radius a. Numerical simulations were performed for values $a = [1, 2, 4, 8, 16, 32, 64]$ mm. The aspect ratio of the ellipse was varied by taking $b/a = \left[0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\right]$. Note that for $b/a = 0$ the strip with crack as given in Fig. 3 is obtained, whereas the case $b/a = 1$ reduces to the strip with circular hole as given in Fig. 4.

In Fig. 5 the strain ratio is plotted against a/ℓ for the five aspect ratios of the ellipse. Only simulations with the ε-RA approach are shown since singularities remain in the u-RA approach for $b/a = 0$ (see Fig. 3) and the differences between the Peerlings approach and the ε-RA approach are marginal. It can be seen that in the small-size limit all five geometries tend to predict roughly the same strain ratio. More specifically, approximately the same maximum non-local equivalent strain is found for small specimens, irrespective of the aspect ratio of the ellipse. For the large-size limit the size effect curves approach asymptotes, whereby the limit cases are

- a horizontal asymptote for the case $b/a = 1$, i.e. a circular hole. This implies that no size effect is present for the large-size range of this geometry;
- an inclined asymptote with slope $\approx -\frac{1}{2}$ for the case $b/a = 0$, i.e. a crack. This implies that a size effect is present for the large-size range. In particular, for larger specimen the smoothing effects of the gradients become negligible, and the solutions converge towards those of linear elastic fracture mechanics (LEFM).

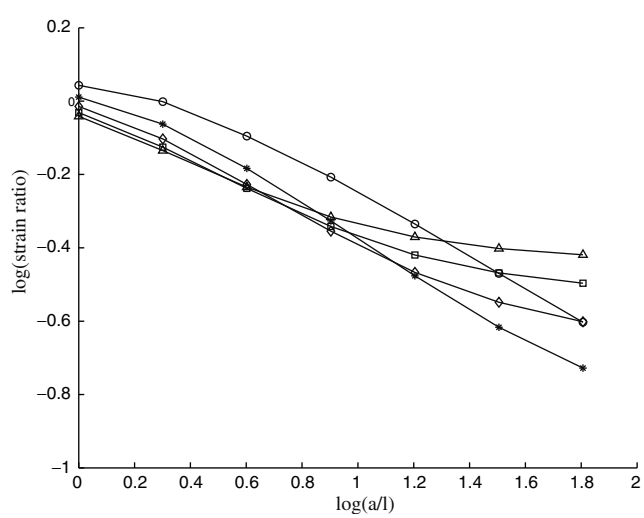

Fig. 5. Strip with ellipsoidal hole – size effect curves for ε-RA approach and different ellipse aspect ratios: $b/a = 0$ (circles), $b/a = \frac{1}{4}$ (asterisks), $b/a = \frac{1}{2}$ (diamonds), $b/a = \frac{3}{4}$ (squares) and $b/a = 1$ (triangles).

6.3. Strip with wedge-shaped notches

Finally, a strip with wedge-shape notches is analysed as studied earlier in [58]. The geometry and loading conditions are shown in Fig. 6. Symmetry conditions are imposed on the left and bottom edges, and a prescribed displacement is imposed at the right edge. The strip has a horizontal dimension $2L$ and a vertical dimension L, but only the top right quarter is modelled. The notch dimensions are given by D. Different specimen sizes are simulated, given by $L = [4, 8, 16, 32, 64, 128, 256]$ mm. The purpose of this test is to investigate the behaviour of specimens where the notch dimensions do or do not scale with the specimen dimensions: a distinction is made between proportional notches (where $D = \frac{1}{4}L$) and constant notches (where $D = 1$ mm for all specimen sizes L). Again, only the ε-RA approach is used.

In Fig. 7 the strain ratio at the notch is plotted against L/ℓ for specimens with constant notch and with propor-

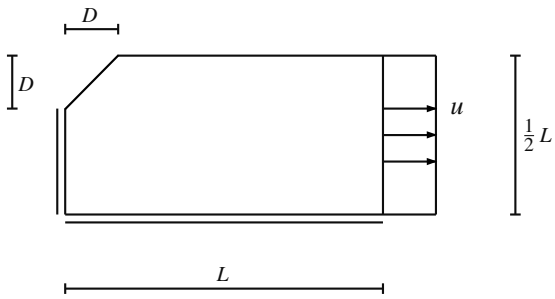

Fig. 6. Strip with wedge-shaped notches – geometry and loading conditions.

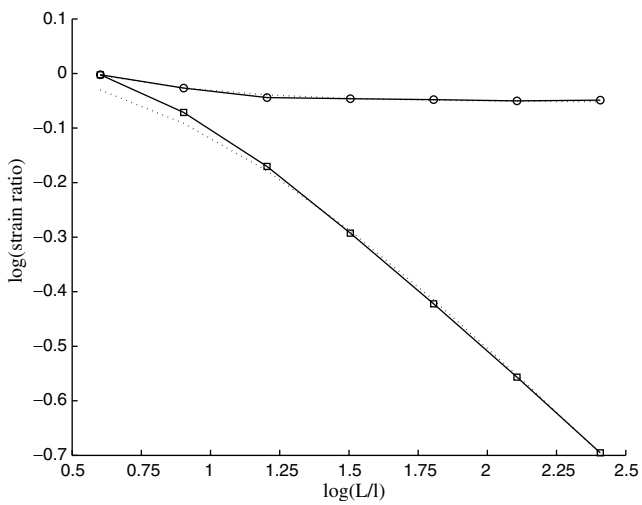

Fig. 7. Strip with wedge-shaped notches – size effect curves for ε-RA approach: constant notches (circles) and proportional notches (squares). Dotted lines denote fits of Multi-Fractal Scaling Law to the constant notch curve and of Size-Effect Law to the proportional notch curve.

tional notch. The constant notch curve renders a horizontal asymptote in the large-size range, which implies that the size effect disappears for large specimens. Note that in this case the notch dimension D and the material length parameter ℓ have the same magnitude, so that neither of them can be neglected. On the other hand, the proportional notch curve attains an inclined asymptote, the slope of which is $-\frac{1}{2}$.

It is interesting to compare the results presented in Fig. 7 with scaling laws that have been proposed in the literature. Although these scaling laws are normally formulated for a so-called nominal stress (which equals the load at which the structure fails divided by some characteristic structural area), the same trends are observed as for the strain ratio defined in Eq. (18). A particular scaling law that has been proposed for the scaling of unnotched specimens is the Multi-Fractal Scaling Law (MFSL) [77], whereby

$$\text{MFSL}: \quad \sigma_{\text{nom}} = A_1\sqrt{1+\frac{A_2}{L}} \tag{19}$$

with σ_{nom} the nominal stress, and A_1 and A_2 are constants that can be fitted. Instead of the nominal stress, however, the strain ratio of Eq. (18) is used. Following the results of Fig. 7, the fitting procedure yields $A_1 = 0.886$ and $A_2 = 1.017\ \text{mm}^{-1}$. The size effect curve according MFSL is also plotted in Fig. 7 and it can be seen that an excellent fit is obtained.

Remark 6. It could be argued that MFSL is not suitable for specimens with a constant notch. However, the simultaneous presence of gradients and a constant notch has the same effect as the absence of a notch: the gradients smoothen any singularity that may arise due to the notch, and in the large-size range there is no difference between a constant notch or an absent notch.

Another scaling law that has been proposed in the literature is the so-called Size Effect Law (SEL) [78], that takes the following format for proportionally notched specimens:

$$\text{SEL}: \quad \sigma_{\text{nom}} = \frac{B_1}{\sqrt{1+L/B_2}}. \tag{20}$$

Again, the nominal stress is replaced by the strain ratio. B_1 and B_2 are constants that after fitting the proportional notch curve read $B_1 = 1.136$ and $B_2 = 8.271\ \text{mm}^{-1}$. Fig. 7 contains also the SEL curve, and again the fit is excellent.

7. Conclusions

In this paper finite element solution strategies have been presented for gradient elasticity. The original fourth-order equations of gradient elasticity are split into two sets of second-order equations. The first set consists of the classical equations of elasticity and in the second set Helmholtz equations are applied to the displacements (the displacement-based Ru–Aifantis or u-RA approach), to the strains (the strain-based Ru–Aifantis or ε-RA approach), to the stresses (the stress-based Ru–Aifantis or σ-RA approach) or to a strain invariant (the Peerlings approach). This two-step approach has the crucial advantage that $\mathscr{C}^0$-continuous interpolations suffice for discretisation, which renders a finite element implementation straightforward. All governing equations including the variationally consistent boundary conditions were presented. It was shown that the ε-RA and σ-RA approaches yield different results in case stiffness jumps are present in the analysed specimen; otherwise, their results are the same.

In a first series of numerical tests, the removal of singularities that occur in classical elasticity was verified. It was found that singularities persist in some components of the strain if the u-RA approach is taken. Conversely, all singularities are removed if the ε-RA approach or the Peerlings approach is used. It is concluded that the homogeneous natural boundary conditions in terms of strain derivatives are sufficient in order to remove all singularities from the strain field.

A second series of test was concerned with the description of a size effect, which was understood here as the capability of the models to predict a changing mechanical behaviour of

the structure if the dimensions of the structure are scaled. All three approaches are capable to describe size effects, although the two strain-based approaches (ε-RA and Peerlings) predict a more significant size effect than the displacement-based approach (u-RA). Finally, the performance of the ε-RA approach is verified against two scaling laws from the literature: Multi-Fractal Scaling Law (MFSL) and Size Effect Law (SEL). The ε-RA approach corresponds excellently with MFSL for specimens in which the imperfection does not scale proportionally with the specimen dimensions. Similarly, an excellent fit is obtained between the ε-RA results and SEL for specimens in which the imperfections do scale proportionally with the specimen dimensions.

The presented examples have shown the power and versatility of gradient elasticity. In combination with the suggested $\mathscr{C}^0$-continuous finite element implementation, the advantages of gradient elasticity can be applied to a much wider range of problems.

Appendix A. Strain energy formulation of fourth-order gradient elasticity

In this appendix the continuum equations underlying the fourth-order gradient elasticity model of Eq. (1) are summarised briefly. The strain energy density $\mathscr{W}$ is given by

$$\mathscr{W} = \frac{1}{2}\varepsilon_{ij}C_{ijkl}\varepsilon_{kl} + \frac{1}{2}\ell^2\varepsilon_{ij,m}C_{ijkl}\varepsilon_{kl,m}, \tag{21}$$

where the constitutive tensor C is given by

$$C_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \mu\delta_{ik}\delta_{jl} + \mu\delta_{il}\delta_{jk} \tag{22}$$

and λ and μ are the Lamé constants. A zeroth-order stresses σ^0 can be defined as

$$\sigma^0_{ij} = \frac{\partial\mathscr{W}}{\partial\varepsilon_{ij}} = C_{ijkl}\varepsilon_{kl}, \tag{23}$$

whereas the higher-order stresses τ is written as

$$\tau_{ijm} = \frac{\partial\mathscr{W}}{\partial\varepsilon_{ij,m}} = \ell^2 C_{ijkl}\varepsilon_{kl,m}. \tag{24}$$

The equilibrium equations are then written in terms of a Cauchy stress σ that takes the following format:

$$\sigma_{ij} = \sigma^0_{ij} - \tau_{ijm,m} = C_{ijkl}(\varepsilon_{kl} - \ell^2\varepsilon_{kl,mm}), \tag{25}$$

which is identical to Eq. (1).

Appendix B. Governing equations of the bar problem

The field equations, boundary conditions and interface conditions for all relevant formulations of the bar problem are listed below.

B.1. Original fourth-order model

The one-dimensional version of Eq. (1) reads

$$u^{\mathrm{g}}_{,xx} - \ell^2 u^{\mathrm{g}}_{,xxxx} = 0, \tag{26}$$

which holds for $-L < x < 0$ and $0 < x < L$ separately. Writing the two solutions u^{g}_1 and u^{g}_2 with two sets of constants gives

$$-L < x < 0: \quad u^{\mathrm{g}}_1 = A_1 x + A_2 + A_3\exp\left(\frac{x}{\ell}\right) + A_4\exp\left(-\frac{x}{\ell}\right), \tag{27a}$$

$$0 < x < L: \quad u^{\mathrm{g}}_2 = A_5 x + A_6 + A_7\exp\left(\frac{x}{\ell}\right) + A_8\exp\left(-\frac{x}{\ell}\right). \tag{27b}$$

The boundary conditions are taken as prescribed values for the displacement and its second derivatives; for interface conditions continuity is required of all four quantities given in Table 1. In particular,

$$x = -L: \quad u^{\mathrm{g}}_1 = 0 \Rightarrow -A_1 L + A_2 + A_3\exp(-\beta) + A_4\exp(\beta) = 0, \tag{28a}$$

$$x = -L: \quad u^{\mathrm{g}}_{1,xx} = 0 \Rightarrow A_3\exp(-\beta) + A_4\exp(\beta) = 0, \tag{28b}$$

$$x = 0: \quad u^{\mathrm{g}}_1 = u^{\mathrm{g}}_2 \Rightarrow A_2 + A_3 + A_4 = A_6 + A_7 + A_8, \tag{28c}$$

$$x = 0: \quad u^{\mathrm{g}}_{1,x} = u^{\mathrm{g}}_{2,x} \Rightarrow \ell A_1 + A_3 - A_4 = \ell A_5 + A_7 - A_8, \tag{28d}$$

$$x = 0: \quad t_1 = t_2 \Rightarrow E_1 A_1 = E_2 A_5, \tag{28e}$$

$$x = 0: \quad m_1 = m_2 \Rightarrow E_1(A_3 + A_4) = E_2(A_7 + A_8), \tag{28f}$$

$$x = L: \quad u^{\mathrm{g}}_2 = \bar{u} \Rightarrow A_5 L + A_6 + A_7\exp(\beta) + A_8\exp(-\beta) = \bar{u}, \tag{28g}$$

$$x = L: \quad u^{\mathrm{g}}_{2,xx} = 0 \Rightarrow A_7\exp(\beta) + A_8\exp(-\beta) = 0, \tag{28h}$$

where $\beta = L/\ell$.

B.2. u-RA approach

For the first step of the staggered approaches the one-dimensional format of Eq. (6) is used, which leads to a piecewise linear response for the classical displacement field u^{c}, i.e.

$$-L < x < 0: \quad u^{\mathrm{c}}_1 = B_1 x + B_2, \tag{29a}$$

$$0 < x < L: \quad u^{\mathrm{c}}_2 = B_5 x + B_6. \tag{29b}$$

The classical displacement field u^{c} must fulfill the following boundary conditions and interface conditions:

$$x = -L: \quad u^{\mathrm{c}}_1 = 0 \Rightarrow -B_1 L + B_2 = 0, \tag{30a}$$

$$x = 0: \quad u^{\mathrm{c}}_1 = u^{\mathrm{c}}_2 \Rightarrow B_2 = B_6, \tag{30b}$$

$$x = 0: \quad t_1 = t_2 \Rightarrow E_1 B_1 = E_2 B_5, \tag{30c}$$

$$x = L: \quad u^{\mathrm{c}}_2 = \bar{u} \Rightarrow B_5 L + B_6 = \bar{u}. \tag{30d}$$

The Helmholtz equation of the second step is non-homogeneous, which implies that a particular solution and a homogeneous solution for u^{g} must be distinguished. The former equals the solution for u^{c} as given through Eqs. (29a) and (29b), whereas the latter consists of exponential functions. For the u-RA approach the combination of particular solution and homogeneous solutions takes a format identical to that given in Eqs. (27a) and (27b):

$$-L < x < 0: \quad u_1^g = B_1 x + B_2 + B_3 \exp\left(\frac{x}{\ell}\right) + B_4 \exp\left(-\frac{x}{\ell}\right), \tag{31a}$$

$$0 < x < L: \quad u_2^g = B_5 x + B_6 + B_7 \exp\left(\frac{x}{\ell}\right) + B_8 \exp\left(-\frac{x}{\ell}\right), \tag{31b}$$

where the constants have been denoted by B_i instead of A_i due to the different boundary and interface conditions that will be used.

In the weak form of the u-RA approach, the second step is complemented with the following boundary conditions and interface conditions:

$$x = -L: \quad u_1^g = u_1^c \Rightarrow B_3 \exp(-\beta) + B_4 \exp(\beta) = 0, \tag{32a}$$

$$x = 0: \quad u_1^g = u_2^g \Rightarrow B_2 + B_3 + B_4 = B_6 + B_7 + B_8, \tag{32b}$$

$$x = 0: \quad u_{1,x}^g = u_{2,x}^g \Rightarrow \ell B_1 + B_3 - B_4 = \ell B_5 + B_7 - B_8, \tag{32c}$$

$$x = L: \quad u_2^g = u_2^c \Rightarrow B_7 \exp(\beta) + B_8 \exp(-\beta) = 0. \tag{32d}$$

It can be verified that Eqs. (28a) and (28b) are equivalent to Eqs. (30a) and (32a). The same holds for Eqs. (28g) and (28h) in connection to Eqs. (30d) and (32d). Furthermore, Eqs. (28d) and (28e) from the original method correspond identically to Eqs. (32c) and (30c) from the u-RA approach. The remaining expressions, that is Eqs. (28c) and (28f) from the original method versus Eqs. (30b) and (32b) from the u-RA approach, only give identical results in case $E_1 = E_2$.

For the strong form of the u-RA aprpoach it has been suggested to use second (rather than first) derivatives as the additional natural boundary conditions, see Table 1. This implies that Eq. (32c) is then replaced by

$$x = 0: \quad u_{1,xx}^g = u_{2,xx}^g \Rightarrow B_3 + B_4 = B_7 + B_8. \tag{33}$$

However, in this case there is a linear dependence between Eqs. (30b), (32b) and (33). Hence, this formulation cannot be used in this bar problem.

B.3. ε-RA and σ-RA approaches

For the ε-RA approach the sum of particular solution and homogeneous solution of the non-local strain is found as

$$-L < x < 0: \quad \varepsilon_1^g = B_1 + C_3 \exp\left(\frac{x}{\ell}\right) + C_4 \exp\left(-\frac{x}{\ell}\right), \tag{34a}$$

$$0 < x < L: \quad \varepsilon_2^g = B_5 + C_7 \exp\left(\frac{x}{\ell}\right) + C_8 \exp\left(-\frac{x}{\ell}\right), \tag{34b}$$

whereby the new constants have been written as C_i rather than B_i. The following conditions apply in addition to Eqs. (30):

$$x = -L: \quad \varepsilon_{1,x}^g = 0 \Rightarrow C_3 \exp(-\beta) - C_4 \exp(\beta) = 0, \tag{35a}$$

$$x = 0: \quad \varepsilon_1^g = \varepsilon_2^g \Rightarrow B_1 + C_3 + C_4 = B_5 + C_7 + C_8, \tag{35b}$$

$$x = 0: \varepsilon_{1,x}^g = \varepsilon_{2,x}^g \Rightarrow C_3 - C_4 = C_7 - C_8, \tag{35c}$$

$$x = L: \quad \varepsilon_{2,x}^g = 0 \Rightarrow C_7 \exp(\beta) - C_8 \exp(-\beta) = 0. \tag{35d}$$

For this specific geometry, there are no differences between the u-RA approach and the ε-RA approach. Although the constants B_3, B_4, B_7 and B_8 have a different meaning than the corresponding constants C_3, C_4, C_7 and C_8, they nonetheless describe the same strain profile. In particular, the boundary conditions Eqs. (35a) and (35d) are equivalent to Eqs. (32a) and (32d), whereas the interface condition Eq. (35b) corresponds to Eq. (32c). Finally, the equivalence of Eq. (35c) with Eq. (32b) can be appreciated in the context of Eq. (30b).

In going from the ε-RA approach to the σ-RA approach, Eqs. (35b) and (35c) are multiplied with the Young's moduli of the two materials. All other equations remain the same.

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